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# Provable Learning of Linear Energy Models over Function Spaces

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## 1 Introduction

### Abstract

Despite remarkable success of Transformers across various domains, such as language modeling and computer vision, their theoretical understanding is still limited. Their power of in-context learning and generalization to unseen tasks, as well as the training dynamics and interpretability are still not well understood. In this paper, we discuss how a Transformer-like Energy-Based Model can provably learn a distribution over a function space. We consider linear families of energy functionals and show how in the mean-field limit of the model heads, it can recover the parameter with provable approximation and generalization guarantees. Our findings give an insight into the unsupervised training dynamics of Transformers-like models, the role of the positional encodings and the importance of the regularization for the generalization properties. We evaluate our architecture on synthetic and real-world benchmarks to support our theoretical findings.

This paper makes the following contributions:

- **Learning distributions over functions.** We formulate the problem of learning a distribution  $\nu$  over a function space  $\mathcal{F}$  from noisy observations as a regularized optimization problem. We model  $\nu$  as an exponential family with sufficient statistic  $\Phi: \mathcal{F} \rightarrow \mathbb{R}^d$ , so that  $p(f) \propto \exp(-\langle \theta, \Phi(f) \rangle)$ , and state generalization guarantees for several (semi-)parametric choices of  $\Phi$ .
- **Connection to in-context learning.** We draw a connection of this problem to the in-context learning in Transformer-like Energy-Based Models, discuss how it can explain the convergence to the optimal solution, the empirically observed scaling laws, the role of the positional encodings and the importance of the regularization for the generalization properties.
- **Architectural implications and empirical validation.** Guided by the optimization problem, we propose a modification of the Transformer architecture that enforces the structural assumptions of our framework. Experiments on both synthetic and real-world benchmarks demonstrate improvements in size-generalization, interpretability, and computational efficiency.

## 30 2 Related Work

31 Several works have showed that a fixed Transformer can implemet a gradient descent for a simple  
32 class of functions such as linear regression, Bayesian inference, etc.

## 33 3 Problem Setup

34 We are given a manifold  $\mathcal{M}$  equipped with a probability measure  $\mu$ , a function space  $\mathcal{F}$  of functions  
35  $f: \mathcal{M} \rightarrow \mathbb{R}$ , endowed with an *unknown* probability measure  $\nu$  over  $\mathcal{F}$ . We observe a dataset  
36  $\mathcal{D} = \{D^{(1)}, \dots, D^{(m)}\}$  consisting of  $m$  observations  $D^{(j)}$  of functions  $f_1, \dots, f_m \stackrel{\text{i.i.d.}}{\sim} \nu(\mathcal{F})$ , each  
37 evaluated at  $n$  points drawn from  $\mu$ :

$$x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} \mu(\mathcal{M}), \quad \epsilon_{j,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

$$\mathcal{D} = \left\{ \left\{ (x_i, f_j(x_i) + \epsilon_{j,i}) \right\}_{i=1}^n \right\}_{j=1}^m.$$

38 Our goal is to learn the measure  $\nu$  over  $\mathcal{F}$ . Given  $\nu$ , predicting a new function  $f \sim \nu$  from noisy  
39 observations  $\{(x_i, y_i)\}_{i=1}^n$ , where  $y_i = f(x_i) + \epsilon_i$  and  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ , reduces to MAP estimation:

$$\hat{f}_m = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n |y_i - f(x_i)|^2 - \log \nu(f).$$

40 **Connection to ICL in Transformers.** The setup described above is closely related to in-context  
41 learning in Transformers. Given a prompt  $(x_1, y_1), \dots, (x_n, y_n)$ , where  $x_i$  are positional encodings  
42 and  $y_i$  are tokens, the task is to predict the next token at position  $x_{n+1}$ . Each prompt can be viewed  
43 as a noisy observation of a latent function  $f \sim \nu(\mathcal{F})$ , and distinct prompts correspond to distinct  
44 draws from  $\nu(\mathcal{F})$ . The Transformer thus implicitly learns the shared structure across functions, that  
45 is, the measure  $\nu$  itself, and uses it to generalize within a new context.

## 46 4 Learning Energy Functionals In Theory And Practice

47 In this section, we derive an optimization problem to address the goal described in Section 3.

48 **Assumption 4.1.** The function domain  $\mathcal{M}$  is a smooth Riemannian manifold embedded in a Euclidean  
49 space  $\mathcal{M} \subset \mathbb{R}^d$  equipped with a Riemannian metric  $\langle \cdot, \cdot \rangle$  and a probability measure  $\mu$  over it.

50 **Assumption 4.2** (Gibbs distribution). We assume that the probability measure  $\nu$  over  $\mathcal{F}$  is a Gibbs  
51 distribution induced by a convex energy functional  $E_\theta : \mathcal{F} \rightarrow \mathbb{R}^+$  and a reference measure (**TODO:**  
52 **define**)  $\nu_0$  on  $\mathcal{F}$ :

$$\nu(f) = \frac{1}{Z_\theta} \exp(-E_\theta(f))$$

53 where  $Z_\theta = \int_{f \in \mathcal{F}} \exp(-E_\theta(f)) \nu_0(df)$  is a normalizing constant and  $E_\theta$  is a family of functionals  
54 parametrized by  $\theta \in \Theta$ .

### 55 4.1 Function Extrapolation

56 First, let us see how knowing the energy functional enables function extrapolation. Suppose we  
57 observe  $D = \{(x_i, y_i)\}_{i=1}^n$  with  $y_i = f(x_i) + \epsilon_i$ ,  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ , and wish to predict  $y_{m+1} =$   
58  $f(x_{m+1})$  at a new point  $x_{m+1}$ . Given the Gibbs prior  $p(f) \propto \exp(-E(f))$ , we recover  $f$  via MAP  
59 estimation:

$$\hat{f}_n = \arg \max_{f \in \mathcal{F}} p(D | f) p(f) = \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{\sigma^2} \sum_{i=1}^n |y_i - f(x_i)|^2 + E(f).$$

60 This estimation will have a generalization guarantee of extrapolating  $f$  to a new point  $x_{n+1} \sim \mu$  via  
61 the following theorem:

62 **Theorem 4.3.** *[TODO: add theorem]*

For example, assuming that our function is  $s$ -smooth on  $\mathcal{M} \subset \mathbb{R}^d$  then Theorem 4.3 recovers the standard rate  $O(n^{-2s/(2s+d)})$  of the minimax rate of estimation of a smooth function from  $n$  noisy observations.

## 4.2 Likelihood

As we mentioned, the family of the energy functionals  $E_\theta$  depends on some parameter  $\theta$ . Hence, our goal is to learn the parameter  $\theta$  by maximizing the likelihood of the observed data. First, let us consider a simpler case, when we are given full functions  $\mathcal{D} = \{f_1, \dots, f_m\}$ , not just empirical samples  $f_i(x_j) + \epsilon$  and we want to learn the parameter  $\theta$  by maximizing the likelihood of the observed data.

$$\min_{\theta} -\log p_{\theta}(\mathcal{D}) = \frac{1}{m} \sum_{i=1}^m E_{\theta}(f_i) + \log Z(\theta)$$

where  $Z(\theta) = \int_{f \in \mathcal{F}} \exp(-E_{\theta}(f)) \nu(df)$  is a normalizing constant,  $\nu$  base reference measure on  $\mathcal{F}$ .

Now, back to the original problem, where each function  $f$  is given by its values:  $D = \{(x_i, y_i)\}_{i=1}^n$  such that  $y_i = f(x_i) + \epsilon_i$ ,  $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ . So given an observation  $D^{(i)} = \{(x_j, y_j)\}_{j=1}^n$  our optimization problem turns into

$$\min_{\theta} \mathcal{L}(\theta) = -\frac{1}{m} \sum_{i=1}^m \log p_{\theta}(D^{(i)}) + \log Z(\theta) \quad (1)$$

$$\text{where } \log p_{\theta}(D^{(i)}) = \log \int_{f \in \mathcal{F}} \exp(-\log p(D^{(i)} | f) - E_{\theta}(f)) \nu(df)$$

Unfortunately, in practice each term  $\log p_{\theta}(D^{(i)})$  is intractable. However, we can approximate it by a bilevel optimization problem.

## 4.3 Bilevel Optimization Via Masking

In this section, we discuss how to approximate our intractable convex optimization problem (1) by a bi-level non-convex optimization problem via masking approach. Fix a parameter  $\theta$  and a dataset  $D = \{(x_j, y_j)\}_{j=1}^n$ . Define  $L_{\theta}(f, D) := \ell(D, f) + \lambda E_{\theta}(f)$ , where  $\ell(D, f)$  denotes the empirical negative log-likelihood and  $E_{\theta}$  is an energy functional.

Note, for a fixed  $\theta$  the minimizing  $L_{\theta}(f, D)$  with respect to  $f$  is unique:  $\hat{f}_{\theta} = \arg \min_{f \in \mathcal{F}} L_{\theta}(f, D)$ . Moreover, the posterior distribution  $p_{\theta}(f | D) \propto \exp(-L_{\theta}(f, D))$  concentrates in probability around  $\hat{f}_{\theta}$  as either  $n \rightarrow \infty$  or  $\sigma^2 \rightarrow 0$ . In particular, for any neighborhood  $U$  of  $\hat{f}_{\theta}$  in  $\mathcal{F}$ ,  $p_{\theta}(U | D) \rightarrow 1$ . Using a second-order Taylor expansion,

$$E_{\theta}(f) = E_{\theta}(\hat{f}_{\theta}) + \frac{1}{2} \langle f - \hat{f}_{\theta}, H_{\theta}(f - \hat{f}_{\theta}) \rangle + o(\|f - \hat{f}_{\theta}\|_X^2).$$

where Hessian operator  $H_{\theta} := \nabla^2 E_{\theta}(\hat{f}_{\theta})$  is the second-order derivative of the energy functional  $E_{\theta}$ , we can approximate the intractable integral as:

$$\int_X \exp(-\log p(D | f) - E_{\theta}(f)) \nu(df) \approx \exp(-\log p(D | \hat{f}_{\theta}) - E_{\theta}(\hat{f}_{\theta})) (\det H_{\theta})^{-1/2}. \quad (2)$$

Substituting the Laplace approximation (2) into the negative log marginal likelihood yields

$$-\log p_{\theta}(D) \approx \log p(D | \hat{f}_{\theta}) + E_{\theta}(\hat{f}_{\theta}) + \frac{1}{2} \log \det H_{\theta} + \log Z(\theta) \quad (3)$$

up to additive constants independent of  $\theta$ . Using this approximation, we can turn the intractable convex optimization problem (1) into a tractable bilevel non-convex surrogate of it. Given the

88 collection of datasets  $\mathcal{D} = \left\{ \left( D^{(i)}, D_{\text{mask}}^{(i)} \right) \right\}_{i=1}^m$ . Substituting this into Equation (1) we have

$$\min_{\theta \in \Theta} \mathcal{L}(\theta) = \frac{1}{m} \sum_{i=1}^m \log p \left( D_{\text{mask}}^{(i)} \mid \hat{f}_{\theta}^{(i)} \right) + E_{\theta} \left( \hat{f}_{\theta}^{(i)} \right) + \frac{1}{2} \log \det H_{\theta}^{(i)} - \log Z(\theta) \quad (\text{outer problem})$$

$$\text{where } \hat{f}_{\theta}^{(i)} = \operatorname{argmin}_{f \in \mathcal{F}} \log p \left( D^{(i)} \mid f \right) + E_{\theta}(f) \quad (\text{inner problem}) \quad (4)$$

89 The inner problem has parameter  $\theta$  fixed and find the function  $\hat{f}_{\theta}^{(i)}$  that minimizes the loss + energy  
 90 functional. The outer problem then finds the parameter  $\theta$  yields the best prediction of  $\hat{f}_{\theta}^{(i)}$  on the  
 91 masked dataset  $D_{\text{masked}}^{(i)}$ . Both problems are convex on their own but not jointly convex. However, we  
 92 can solve it by alternating gradient descent with a following guarantee:

93 **Theorem 4.4.** *[TODO: add theorem]*

94 **The Inner Problem.** For a dataset  $D = \{(x_i, y_i)\}$  we have

$$\hat{f}_{\theta} = \operatorname{argmin}_f \log p(D \mid f) + E_{\theta}(f) = \operatorname{argmin}_f \|f(x_i) - y_i\|_2^2 + \langle \theta, \Phi(f) \rangle$$

95 Note, that due to convexity, this can be solved by subgradient descent on  $f$  with a step size  $\eta > 0$ :

$$f^{(t+1)} = f^{(t)} - \eta \left[ 2 \sum_{i=1}^n (f^{(t)}(x_i) - y_i) \delta_{x_i} + \nabla_f E_{\theta}(f^{(t)}) \right]$$

96 where  $\nabla_f E_{\theta}(f^{(t)})$  is a (sub)gradient of the energy functional.

97 **The Outer Problem.**

## 98 5 Families Of Energy Functionals

99 Here we look closer at specific linear families of energy functionals  $E_{\theta} = \langle \theta, \Phi(f) \rangle$  defined by the  
 100 statistic extractor  $\Phi$ . We discuss what inductive bias each choice of  $\Phi$  induces, what generalization  
 101 properties it can achieve and how a transformer forward pass can be similar to a (sub)gradient descent  
 102 for the inner problem, and transformer training can be seen as a gradient descent on the outer problem.

103 **Notation** Let  $\mathcal{Q} \subset \mathbf{S}_+^d$  be *[TODO: define]*. Given a PSD matrix  $Q$ , let  $\Delta_Q$  denotes the  
 104 anisotropic Laplacian operator:  $\Delta_Q f := \nabla \cdot (Q \nabla f)$ . Total variation norm is defined as  
 105  $\|g\|_{\mathcal{M}} := \int_{x \in \mathcal{M}} \|g(x)\|_{\mathcal{M}} dx$  where  $g$  *[TODO: can be a Radon measure.]*

106 **Anisotropic Sobolev regularization.** Let  $\theta$  be a measure over  $\mathcal{Q}$ . Let statistic  $\Phi : H^1(M) \rightarrow \mathbb{R}^k$   
 107 be defined as  $\Phi(f)(Q) := \int_{x \in \mathcal{M}} \|\nabla f(x)\|_Q^2 dx$ . Then:

$$E_{\theta}(f) = \int_{Q \in \mathcal{Q}} \Phi(f)(Q) d\theta(Q) \quad \nabla_f E_{\theta}(f) = \int_{Q \in \mathcal{Q}} \Delta_Q f d\theta(Q)$$

108 **Anisotropic second-order TV** Define statistic  $\Phi : X \rightarrow \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+)$  as  $\Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}$ .  
 109 Then:

$$E_{\theta}(f) = \int_{\mathcal{Q}} \|\Delta_Q f\|_{\mathcal{M}} d\theta(Q) \quad \nabla_f E_{\theta}(f) = \int_{\mathcal{Q}} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

110 This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

111 **Spectral energy features.** Let  $\{\psi_k\}$  be eigenfunctions of a Laplacian on  $M$ . Define  $\Phi : L^2(M) \rightarrow$   
 112  $\ell^1$ ,  $\Phi_k(f) := |\langle f, \psi_k \rangle|^2$ .

113 The corresponding regularizer weights energy by frequency and induces a spectral bias.

## 114 6 Connection to Transformers And Its Implications

115 In most of the examples above, the key step of implementing a (sub)gradient descent for the inner  
 116 problem is to compute the discrete (anisotropic) Laplacian  $\Delta_Q f$  for some PSD matrix  $Q$ . The original  
 117 Laplacian  $\Delta_Q$  can be approximated by its discrete version  $\hat{\Delta}_Q \in \mathbb{R}^{n \times n}$  computed on a set of points  
 118  $x_1, \dots, x_n \in \mathcal{M}$ :

$$\hat{\Delta}_Q = I - D^{-1/2} W D^{-1/2} \quad \text{where}$$

$$W_{ij} = \exp\left(\frac{-\langle x_i - x_j, \Pi_{x_i} Q^\dagger \Pi_{x_i}(x_i - x_j) \rangle}{\epsilon}\right) \quad \text{and} \quad D_{ii} = \sum_{j=1}^n W_{ij}.$$

119 Even though the attention mechanism is not exactly the same as the discrete Laplacian, it is similar to  
 120 it in a few ways. The attention mechanism is a way to compute the weighted average ..... The discrete  
 121 Laplacian is a way to compute the weighted average of the features of the points in the neighborhood  
 122 of a given point. Consider a transformer-like architecture where the softmax attention is replaced  
 123 with 2 different types of heads per layer:

- 124 • 1 **linear** head:  $y_{t+1} = (I)$
- 125 •  $H$  **diffusion** heads each parametrized by some  $Q_h \in \mathcal{Q}$  and  $\theta_h \in \mathbb{R}^+$ :  $\bar{y}^{(t+1)} = I -$   
 126  $\eta \theta_h \Delta_{Q_h} \bar{y}^{(t)}.$

127 Then the forward pass of

